Advanced machine learning

Theory and generative models

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Overview

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2. Results from last week's competition
3. Convolutions and Fourier transform
   - Discrete convolutions
   - Discrete Fourier transform
   - Graph convolutions and Fourier transform
4. GNNs and probabilistic models
5. Generative models
   - Traditional methods
   - Variational autoencoders
   - Generative adversarial networks (GANs)
   - Evaluating graph generation
6. Exercises

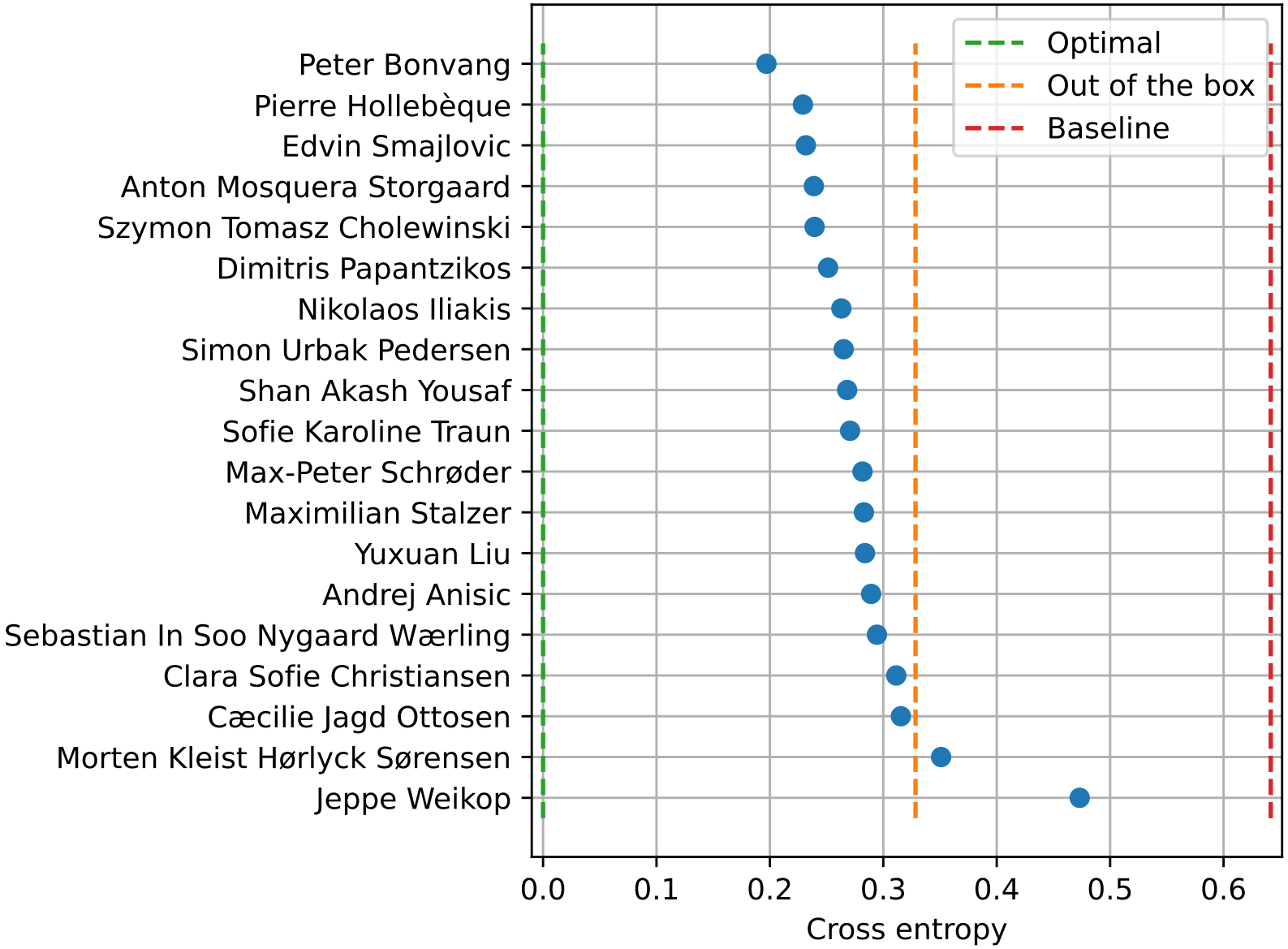
Module overview

Module 3: Graph models

1. *Graphs & node embeddings*
   Reading: Hamilton ch. 1–4
   Exercise 1
2. *Graph neural networks*
   Reading: Hamilton ch. 5–6
   Exercise 2
3. *Theory & generative models*
   Reading: Hamilton ch. 7–9
   Exercise 3
4. *Mini project 3*

Results from last week's competition

Last week's competition



Convolutions and Fourier transform

Convolutions and Fourier transform: Discrete convolutions

Convolution

- Convolution of discrete time series

$$y[n] = (x * h)[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]$$

- Convolution of continuous signals

$$g(\boldsymbol{x}) = (f * h)(\boldsymbol{x}) = \int_{\mathbb{R}^d} f(\boldsymbol{y})h(\boldsymbol{x}-\boldsymbol{y})\mathrm{d}\boldsymbol{y}$$

We only look at discrete convolutions in this course.

Causal filter

- When the filter coefficients are zero for $n < 0$, i.e. $h[n] = 0 \ \forall \ n < 0$, we say the filter is causal.
  This means that the filter does not respond before the input signal is applied.

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- We can then write the convolution as

$$y[n] = h[0]x[n] + h[1]x[n-1] + h[2]x[n-2] + \dots$$

  This shows how the convolution is a weighted linear combination of delayed versions of the input signal.

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$$y[n] = h[0]x[n] + h[1]x[n-1] + h[2]x[n-2] + \dots$$

  This shows how the convolution is a weighted linear combination of delayed versions of the input signal.
- If we define a unit delay operator

$$Dx[n] = x[n-1]$$

  we can further write it as

$$y[n] = \big(h[0] + h[1]D + h[2]D^2 + \cdots\big)x[n]$$

  This is a nice parallel to graph convolutions as we will see later.

Circular convolution

- Circular discrete-time convolution

$$y[n] = (x \circledast h)[n] = \sum_{k=0}^{N-1} x[k] \cdot h[(n-k) \mod N]$$

- We can define a (Toeplitz structured) filter matrix

$$\boldsymbol{H} = \begin{bmatrix} h[0] & h[N-1] & h[N-2] & \cdots & h[1] \\ h[1] & h[0] & h[N-1] & \cdots & h[2] \\ h[2] & h[1] & h[0] & \cdots & h[3] \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ h[N-1] & h[N-2] & h[N-3] & \cdots & h[0] \end{bmatrix} \quad \text{i.e. } H_{kn} = h[(n-k) \mod N]$$

a compute the convolution as

$$\boldsymbol{y} = \boldsymbol{H}\boldsymbol{x}$$

Convolutions and Fourier transform: Discrete Fourier transform

Discrete Fourier transform

- The discrete Fourier transform is defined as

$$\tilde{x}[k] = \sum_{n=0}^{N-1} x[n] \overbrace{e^{-i\frac{2\pi k}{N}n}}^{\cos\left(\frac{2\pi kn}{N}\right) - i\sin\left(\frac{2\pi kn}{N}\right)} = \sum_{n=0}^{N-1} x[n] \underbrace{\omega_N^{k\cdot n}}_{\omega_N = e^{-i\frac{2\pi}{N}}}$$

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- We can define a DFT basis matrix

$$\boldsymbol{U}^H = \frac{1}{\sqrt{N}} \begin{bmatrix} \omega_N^{0\cdot 0} & \omega_N^{0\cdot 1} & \cdots & \omega_N^{0\cdot (N-1)} \\ \omega_N^{1\cdot 0} & \omega_N^{1\cdot 1} & \cdots & \omega_N^{1\cdot (N-1)} \\ \vdots & \vdots & \ddots & \vdots \\ \omega_N^{(N-1)\cdot 0} & \omega_N^{(N-1)\cdot 1} & \cdots & \omega_N^{(N-1)\cdot (N-1)} \end{bmatrix} \qquad \text{i.e. } U_{kn}^H = \omega_N^{k\cdot n}$$

The DFT basis is a unitary matrix, $\boldsymbol{U}^H \boldsymbol{U} = \boldsymbol{I}$.

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- The DFT and inverse DFT can be computed as

$$\tilde{\boldsymbol{x}} = \boldsymbol{U}^H \boldsymbol{x} \qquad \boldsymbol{x} = \boldsymbol{U}\tilde{\boldsymbol{x}}$$

DFT maps a signal onto a basis of complex sinusoids with different frequencies.

Filtering with the DFT

- Convolution in the "time" domain

$$y[n] = (x \circledast h)[n] = \sum_{k=0}^{N-1} x[k] \cdot h[(n-k) \mod N]$$

corresponds to multiplication in the "frequency" domain

$$\tilde{y}[k] = \tilde{x}[k] \cdot \tilde{h}[k]$$

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$$\tilde{y}[k] = \tilde{x}[k] \cdot \tilde{h}[k]$$

- In matrix form

$$\boldsymbol{y} = \boldsymbol{H}\boldsymbol{x} \qquad \Leftrightarrow \qquad \tilde{\boldsymbol{y}} = \tilde{\boldsymbol{h}} \odot \tilde{\boldsymbol{x}}$$

i.e.

$$\boldsymbol{y} = \boldsymbol{U}(\underbrace{\boldsymbol{U}^H\boldsymbol{h}}_{\tilde{\boldsymbol{h}} = \mathrm{DFT}(\boldsymbol{h})} \odot \underbrace{\boldsymbol{U}^H\boldsymbol{x}}_{\tilde{\boldsymbol{x}} = \mathrm{DFT}(\boldsymbol{x})})$$

Proof: Convolution equals DFT multiplication

$$\tilde{y}[k] = \sum_{n=0}^{N-1} y[n]\omega_N^{k \cdot n}$$

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DFT basis

- DFT basis vectors

$$\boldsymbol{u}_k^* = [\omega_N^{k\cdot 0}, \omega_N^{k\cdot 1}, \ldots, \omega_N^{k\cdot (N-1)}]^\top$$

are complex exponentials with normalized frequency $\frac{k}{N}$

$$(\boldsymbol{u}_k^*)_n = \omega_N^{k\cdot n} = e^{-i\frac{2\pi k}{N}n} = \cos\left(\frac{2\pi k}{N}n\right) - i\sin\left(\frac{2\pi k}{N}n\right)$$

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- Come in complex conjugate pairs

$$(\boldsymbol{u}_{N-k}^*)_n = e^{-i\frac{2\pi(N-k)}{N}n} = \overbrace{e^{-i\frac{2\pi N}{N}n}}^{=1} e^{i\frac{2\pi k}{N}n} = e^{i\frac{2\pi k}{N}n} = (\boldsymbol{u}_k)_n \tag{1}$$

i.e. there are two complex conjugate components per frequency.

Convolutions and Fourier transform: Graph convolutions and Fourier transform

Graph convolutions

- For a discrete time signal, we could write a convolution as a weighted sum of delayed versions of the input signal.

$$y[n] = \big(h[0] + h[1]D + h[2]D^2 + \cdots\big)x[n]$$

Here, the shift operator is the unit delay $Dx[n] = x[n-1]$. On a chain graph, it would correspond to moving on to the next node (e.g. clockwise).

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- For a signal propagating on a graph, we can use the adjacency matrix as a "neighborhood shift" operator.
  Thus we can define a graph convolution as:

$$\boldsymbol{y} = \big(h[0]\boldsymbol{I} + h[1]\boldsymbol{A} + h[2]\boldsymbol{A}^2 + \cdots\big)\boldsymbol{x}$$

Eigendecomposition of the cycle graph adjacency matrix

- Eigenvalue decomposition is given by

$$\boldsymbol{A} = \boldsymbol{U}\boldsymbol{\Lambda}\boldsymbol{U}^H$$

where the eigenvectors satisfy

$$\boldsymbol{A}\boldsymbol{u}_k = \lambda_k \boldsymbol{u}_k$$

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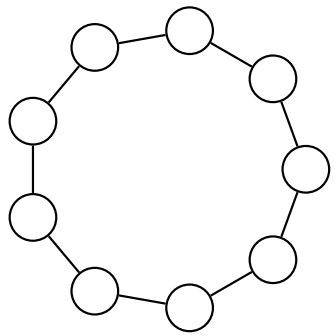

$$\boldsymbol{A} = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

- It turns out that the eigenvectors are equal to the *DFT basis vectors*.
  You will show this in the exercises.

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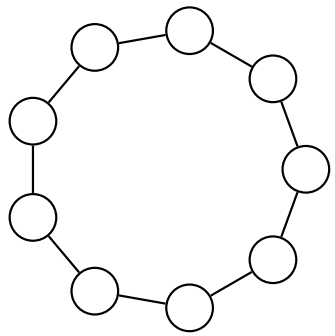

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Note: The eigendecomposition not unique

- Conjugate pairs can be rotated together.
- We can rotate to sine and cosine components, yielding a real matrix $\boldsymbol{U}$.

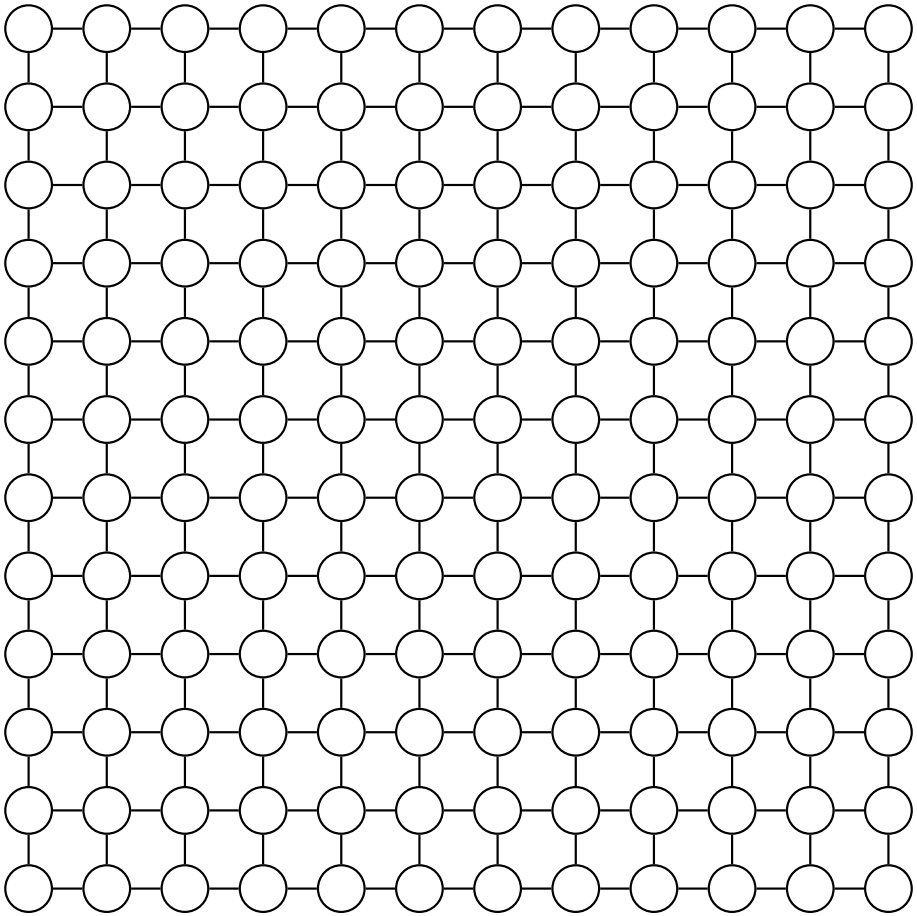
Graph Fourier transform

- We can now define a generalized *graph Fourier transform*, given as the eigendecomposition of the adjacency matrix.

$$\boldsymbol{A} = \boldsymbol{U}\boldsymbol{\Lambda}\,\boldsymbol{U}^H$$

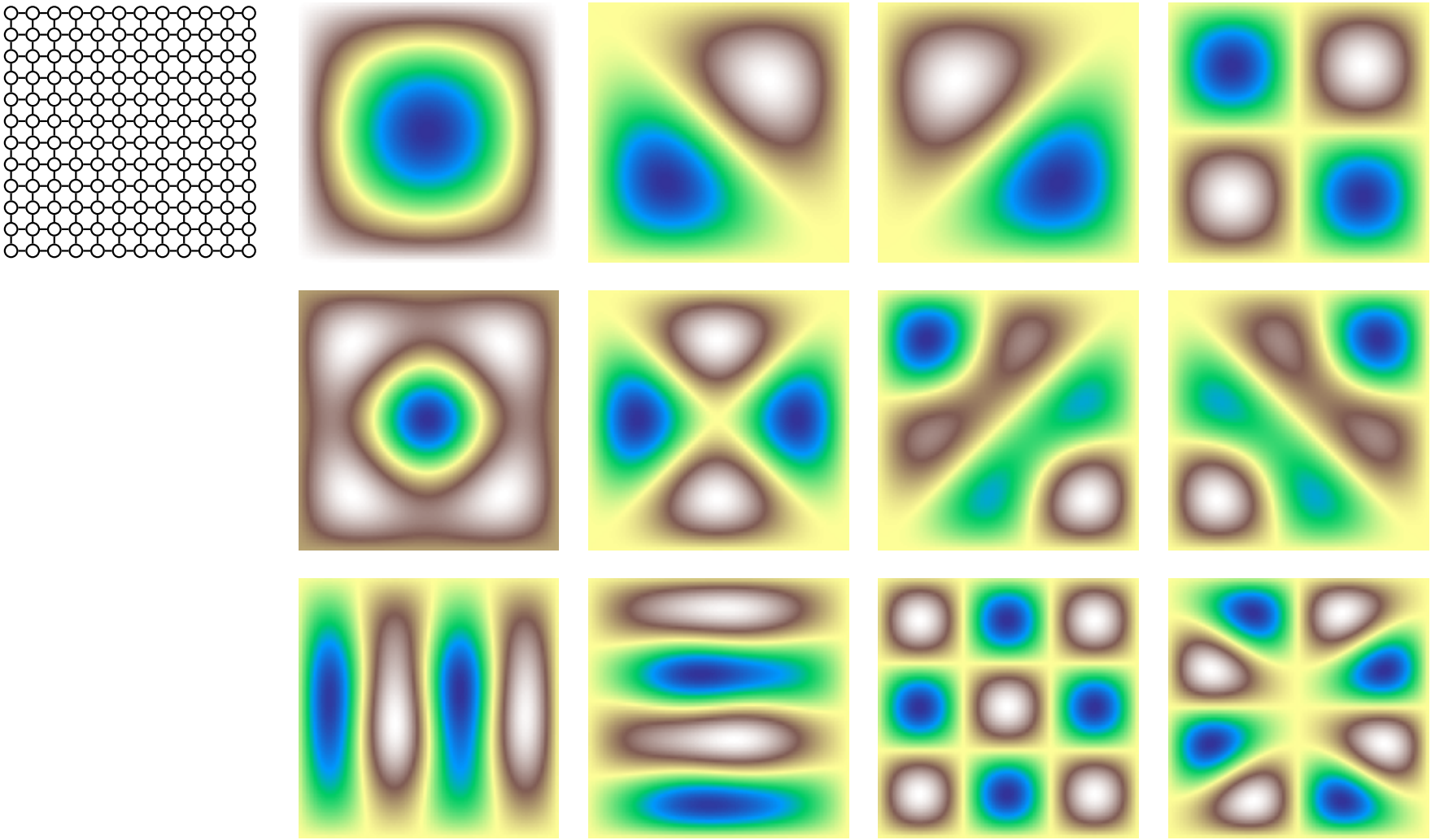
- The basis vectors are no longer *frequencies* but more general *spatial patterns* on the graph vertices.

Example: Grid graph



Example: Spatial components of the grid graph

Twelve "low-frequency" components.



Summary of graph covolution

- A graph convolution is defined as

$$\boldsymbol{y} = \sum_{k=0}^{N} h[k] \boldsymbol{A}^k \boldsymbol{x}$$

- It can equivalently be computed using the graph Fourier transform

$$\boldsymbol{y} = \boldsymbol{U} \left(\sum_{k=0}^{N} h[k] \boldsymbol{\Lambda}^k \right) \boldsymbol{U}^H \boldsymbol{x}$$

where

$$\boldsymbol{A} = \boldsymbol{U} \boldsymbol{\Lambda} \boldsymbol{U}^H$$

GNNs without message passing

- We can now define a graph convolution neural network

$$\overbrace{\boldsymbol{Z}}^{\text{Node-level output features}} = \mathrm{MLP}_{\mathrm{out}}\left(\underbrace{\sum_{k=0}^{N} h[k]\boldsymbol{A}^k}_{\text{Graph convolution}} \overbrace{\mathrm{MLP}_{\mathrm{in}}(\boldsymbol{X})}^{\text{MLP on node input features}}\right)$$

  Computations can also be performed in the Fourier domain.

- No message passing, but information flows through N steps from nodes to neighbors through the graph convolution.

GNNs and probabilistic models

Probabilistic graphical models

- *Probabilistic graphical models:* Represent statistical relationships between random variables using a graph.
- *Approximate inference*: Finding the exact distributions of hidden variables is difficult.
- *Message passing:* Iteratively pass messages between nodes that updates beliefs about node's distribution based on neighbors.
- *Variational approach:* Using a simpler, approximate distribution for each variable.
- *Conjugate models:* Classical variational message passing uses conjugate distributions so updates can be performed exactly.

Kernel mean embedding

- Probability distributions can be embedded in a Hilbert space.
  Hilbert space = A complete vector space with an inner product.
- Mean embedding using a feature map ϕ

$$\boldsymbol{\mu}_x = \int_{\mathbb{R}^d} \phi(\boldsymbol{x}) p(\boldsymbol{x}) \mathrm{d}\boldsymbol{x}$$

  If the feature map is "complex enough" the embedding is injective
- In practice, we do not work directly with the feature map, but define it implicitly using a kernel (inner product).

$$\left\langle \phi(\boldsymbol{x}), \phi(\boldsymbol{x}') \right\rangle = k(\boldsymbol{x}, \boldsymbol{x}')$$

Graphs as graphical models

- A graph defines a Markov random field

$$p\big(\overbrace{\{x_v\}}^{\text{Node features (observed)}}, \underbrace{\{z_v\}}_{\text{Node embedding (latent variables)}}\big) \propto \prod_{v \in \mathcal{V}} \Phi(x_v, z_v) \prod_{(u,v) \in \mathcal{E}} \Psi(z_u, z_v)$$

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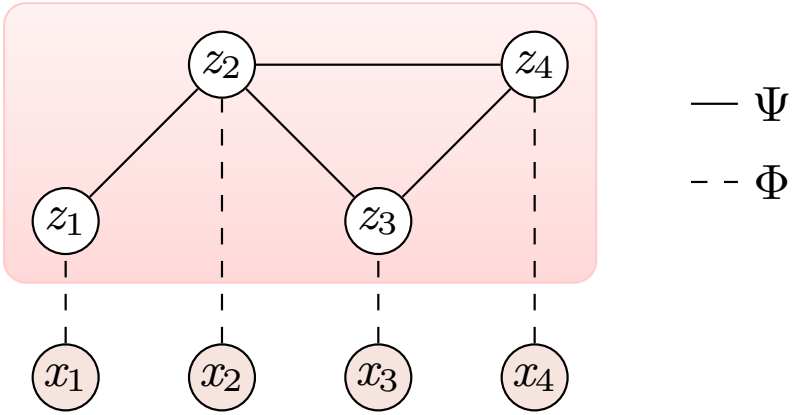
- Example



Nodes could be vector-valued, but I will assume scalars for simplicity.

Mean field variational inference

- Approximate posterior distribution of latent variables

$$p\big(\{z_v\}|\{x_v\}\big) \approx q\big(\{z_v\}\big) = \prod_{v\in\mathcal{V}} q_v(z_v)$$

- Minimize KL to true posterior

$$\mathrm{KL}(q|p) = \int \prod_{v\in\mathcal{V}} q_v(z_v) \log \frac{\prod_{v\in\mathcal{V}} q_v(z_v)}{p\big(\{z_v\}|\{x_v\}\big)} \prod_{v\in\mathcal{V}} \mathrm{d}z_v$$

- Optimum satisifies fixed point equations

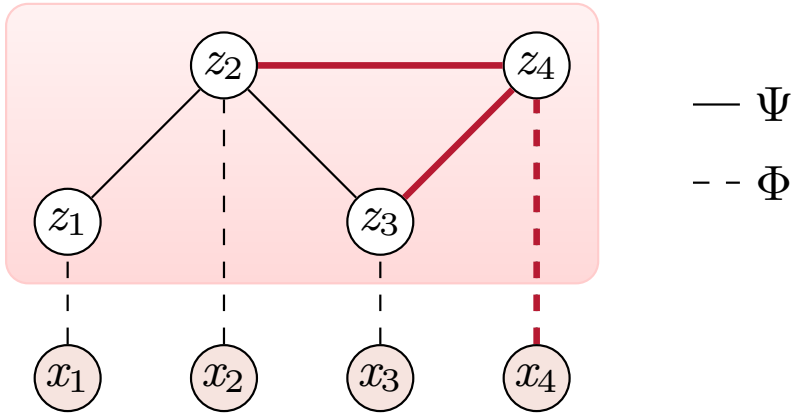
$$\log q_v^{(t+1)}(z_v) = c_v + \log \Phi(x_v, z_v) + \sum_{u\in\mathcal{N}(v)} \int q_u^{(t)}(z_u) \log \Psi(z_u, z_v) \mathrm{d}z_u$$

Mean field variational message passing

- Optimum satisifies fixed point equations

$$\log q_v^{(t+1)}(z_v) = c_v + \log \Phi(x_v, z_v) + \sum_{u \in \mathcal{N}(v)} \int q_u^{(t)}(z_u) \log \Psi(z_u, z_v) \mathrm{d} z_u$$

- Example (node 4)



$$\log q_4^{(t)}(z_4) = c_4 + \log \Phi(x_4, z_4) + \int q_2^{(t)}(z_2) \log \Psi(z_2, z_4) \mathrm{d} z_2 + \int q_3^{(t)}(z_3) \log \Psi(z_3, z_4) \mathrm{d} z_3$$

Mean field VI embedded in Hilbert space

- Suppose we represent $q_v(z_v)$ as mean embeddings

$$\boldsymbol{\mu}_v = \int \phi(z) q_v(z_v) \mathrm{d}\boldsymbol{z}$$

  the fixed point equations can be written as

$$\boldsymbol{\mu}_v^{(t+1)} = \boldsymbol{c} + f\left(\boldsymbol{\mu}_v^{(t)}, x_v, \{\boldsymbol{\mu}_u, \forall u \in \mathcal{N}(v)\}\right)$$

  This looks very much like message passing GNN updates.

- We can now use this to either
  - Define potential functions Φ and Ψ and a kernel $k(\cdot, \cdot)$ and analytically derive updates.
  - Learn embeddings and assume that they *could* correspond to some probabilistic model.

Generative models

Generative models: Traditional methods

Traditional models

- *Erdös-Rényi model*
  Generate each possible edge independently with a fixed probability

$$P(A_{u,v} = 1) = r, \quad \forall u, v \in \mathcal{V}, u \neq v.$$

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- *Stochastic blockmodels*
  1. Specify a categorical distribution $p_1, \ldots p_\gamma$ over blocks $\mathcal{C}_1, \ldots, \mathcal{C}_\gamma$.
  2. Assign each node randomly to a block according to p_i.
  3. Specify probabilities of edges between (and within) each block $r_{i,j}$
  4. Generate edges between nodes in blocks i and j according to

$$P(A_{u,v} = 1) = r_{i,j}, \quad \forall u \in \mathcal{C}_i, v \in \mathcal{C}_j, u \neq v, \forall i, j \in (1, \ldots, \gamma)$$

Possible simplification

$$r_{i,j} = \begin{cases} r_{\text{in}} & i = j \\ r_{\text{out}} & i \neq j \end{cases}$$

Traditional models

- *Preferential attachment*
  1. Start with a seed graph, e.g. a fully connected graph with m_0 nodes (Possibly just a single node)
  2. Iteratively add nodes to the graph and form edges according to

$$P(A_{u,v} = 1) = \frac{d_v^{(t)}}{\sum_{v' \in \mathcal{V}^{(t)}} d_v^{(t)}}$$

     Nodes that are already highly connected are more likely to connect.

  There are also linear and non-linear extensions that allows more detailed control of the distribution.

Applications of traditional methods

- *Generating synthetic benchmark data:* Testing algorithms, validating hypotheses, and studying network phenomena in controlled settings.
- *Creating null models:* If observed properties of a real graph significantly differ from those of the null model, it suggests that the real graph exhibits non-random structural characteristics or underlying processes.

Generative models: Variational autoencoders

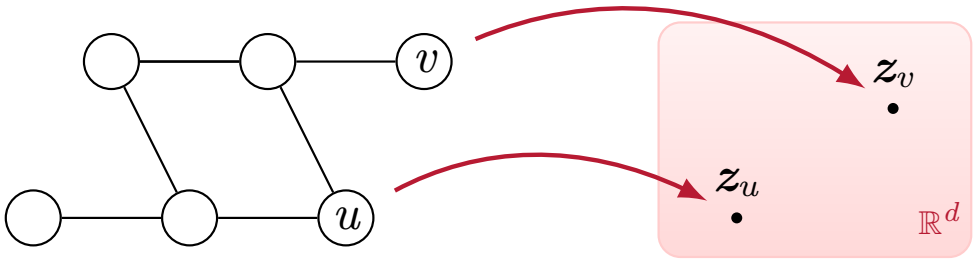
Latent variable models

- We have already worked with a generative model: *Shallow node embedding*.
- Each node is embedded in a space, i.e. is endowed with a latent variable.
- The graph edges are generated according to a pairwise decoder.

Encoder-decoder perspective

Encoder: Maps a vectex into a latent representation.

$$\textsc{enc} : \mathcal{V} \to \mathbb{R}^d$$



Example 1: Learned embedding (lookup table).

Example 2: Message passing graph neural network.

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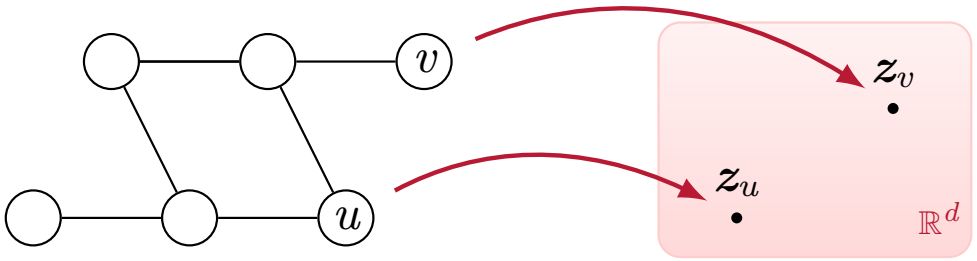

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Pairwise decoder: Maps a pair of latent variables into a node pair statistic.

$$\text{DEC} : \mathbb{R}^d \times \mathbb{R}^d \to \mathbb{S}$$

Example 1: Predict link or non-link, $\mathbb{S} = \{0, 1\}$.

Example 2: Predict a graph-based similarity measure, $\mathbb{S} = \mathbb{R}_+ = \{x \in \mathbb{R} | x \geq 0\}$.

Variational autoencoders

- *Probabilistic encoder:* Takes graph as input and computes parameters of a probabilistic latent embedding.

$$\boldsymbol{Z} \sim \mathcal{N}\left(\boldsymbol{\mu}_\phi(\mathcal{G}), \boldsymbol{\Sigma}_\phi(\mathcal{G})\right)$$

  For example the mean and variance of a Normal distribution.

- *Probabilistic decoder:* Takes a latent representation and produces a conditional distribution over entries of the adjacency matrix.

$$\boldsymbol{A}_{u,v} \sim \text{Bernoulli}\big(p_{u,v}(\boldsymbol{Z})\big)$$

  For example the parameters of independent Bernoulli variables for each possible link.

- *A prior distribution:*

$$p(\boldsymbol{Z}) = \mathcal{N}(\mathbf{0}, \boldsymbol{I})$$

  Typically this is a standard Normal.

Variational autoencoders

- *Probabilistic encoder:* $\boldsymbol{Z} \sim \mathcal{N}\left(\boldsymbol{\mu}_\phi(\mathcal{G}), \boldsymbol{\Sigma}_\phi(\mathcal{G})\right)$
- *Probabilistic decoder:* $\boldsymbol{A}_{u,v} \sim \text{Bernoulli}\big(p_{u,v}(\boldsymbol{Z})\big)$
- *A prior distribution:* $p(\boldsymbol{Z}) = \mathcal{N}(\boldsymbol{0}, \boldsymbol{I})$

Given these components we train the model by maximizing the evidence lower bound (ELBO)

$$\mathcal{L} = \sum_{\mathcal{G} \in \mathcal{D}} \mathbb{E}_{q_\phi(\boldsymbol{Z}|\mathcal{G})}\left[p_\theta(\mathcal{G}|\boldsymbol{Z})\right] - \text{KL}\left[q_\phi(\boldsymbol{Z}|\mathcal{G}) \parallel p(\boldsymbol{Z})\right]$$

Node-level latents

- *Encoder:* Probabilistic node embedding

$$\underbrace{\boldsymbol{\mu}_{\boldsymbol{Z}}, \log \sigma_{\boldsymbol{Z}}}_{\text{Mean and variance for each node}} = \overbrace{\mathrm{GNN}(\boldsymbol{A}, \boldsymbol{X})}^{\text{GNN outputs two vectors per node}}$$

Sample latent node embedding using the reparametrization trick

$$\boldsymbol{Z} = \boldsymbol{\mu}_{\boldsymbol{Z}} + \boldsymbol{\epsilon} \odot \log \sigma_{\boldsymbol{Z}}$$

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- *Decoder:* Probabilistic pairwise decoder compute a binary probability

$$P(A_{u,v} = 1) = \sigma(\boldsymbol{z}_u^\top \boldsymbol{z}_v + \boldsymbol{b}n)$$

Same decoder we used for shallow embeddings.

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Note: The basic node-level VAE has very limited performance, because the latent variable model (decoder) is very weak. This can be improved by using a GNN as decoder also.

Graph-level latents

- *Encoder:* Probabilistic graph embedding

$$\underbrace{\boldsymbol{\mu}_z, \log \sigma_z}_{\text{Mean and variance for the entire graph}} = \overbrace{\mathrm{GNN}(\boldsymbol{A}, \boldsymbol{X})}^{\text{GNN outputs two vectors}}$$

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Note: Two problems:

1. *We need to assume a fixed number of nodes:* This can be handled by assuming a maximum number and masking.
2. *We do not know the order of the nodes:* This can be handled by (heuristic) graph matching to optimize the node order on the fly.

Generative models: Generative adversarial networks (GANs)

Adversarial approaches

- *Generator:* Generates data from a (random) latent variable

$$g_\theta : \mathbb{R}^d \to \mathcal{X}$$

- *Discriminator*: Distinguish between real and generated data

$$d_\phi : \mathcal{X} \to [0, 1]$$

- *Adversarial optimization:* Train the discriminator to distinguish real and generated, while training the generator to fool the discriminator

$$\min_\theta \max_\phi \mathbb{E}_{\underbrace{\boldsymbol{x} \sim p(\boldsymbol{x})}_{\text{Sample from data distribution}}}[\log(1 - d_\phi(\boldsymbol{x}))] + \mathbb{E}_{\underbrace{\boldsymbol{z} \sim p(\boldsymbol{z})}_{\text{Sample from prior}}}[\log d_\phi(g_\theta(\boldsymbol{z}))]$$

GAN for graph generation

- *Generator:* Could be as simple multilayer perceptron.

$$\tilde{\boldsymbol{A}} = \sigma\big(\mathrm{MLP}(\boldsymbol{z})\big)$$

- *Discriminator:* GNN based graph classification.

This approach does not require a specific node ordering: The discriminator is permutation invariant.

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  Measure a distributional distance between generated graphs and real test graphs.
- Distance measure: Total variation

$$d(s_{\text{gen}}, s_{\text{test}}) = \sup_x |P(s_{\text{gen}} \in x), P(s_{\text{test}} \in x)|$$

  Measure the maximum disagreement in probability assignment. For discrete distributions this is simply half the L_1 distance.

Exercises

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A Graph convolutions

B Graph Fourier transform

C Programming exercise (Graph Fourier transform)